Mycelium Electrical Behavior: From Morphological Modeling to Hardware Implementation

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Abstract—This work investigates the morphological and electrical behavior of mycelium by developing a comprehensive hardware framework that links bio-inspired modeling with real-time electronics. We propose a digital twin of mycelium, integrating reaction-diffusion cellular automata (CA) models to simulate fungal growth behaviors and substrate interaction. These CA simulations are enhanced with environmental parameters and optimized through genetic algorithms to match experimental morphological data. We present a systematic transition from general-purpose, GPU-accelerated simulation environments to fixed-point, resource-constrained FPGA implementations, maintaining fidelity while allowing real-time performance. Extracting the morphological structure of mycelium from FPGA, this study explores the use of memristor-based circuits to emulate the adaptive electrical activity observed in fungi. A memristive oscillating framework is developed, enabling hardware-level representation of spiking behaviors and plastic signal propagation in mycelial networks. The integration of biological feedback, real-time control, and non-linear electrical characteristics supports low-power, adaptive systems suitable for neuromorphic and bio-hybrid computing architectures. This research offers a scalable pathway from material modeling to hardware realization, advancing digital twin approaches for engineered living materials.

Index Terms—Mycelium Modeling, Cellular Automata, Reaction-Diffusion, FPGA, Digital Twins, Memristors, Bio-hybrid Systems, Neuromorphic Computing

I. Introduction

The study of engineered living materials (ELMs), particularly mycelium-based substrates, has opened new avenues for bio-hybrid and neuromorphic systems [1], [2]. Mycelium, the filamentous root structure of fungi, exhibits rich morphological evolution and adaptive electrical activity, making it an ideal candidate for unconventional computing and environmental sensing applications [3], [4]. Understanding and emulating these behaviors in silico and in hardware is essential for building responsive and self-organizing systems that bridge biological growth and electronic computation [5].

This paper presents a unified framework for implementing the morphological and electrical behaviors of mycelium within a digital twin architecture. The proposed framework integrates biologically inspired reaction-diffusion cellular automata (CA) models enabling both accurate simulation and hardware compatibility. The CA model simulates hyphal tip extension, branching, and substrate interaction under varying environmental conditions, while a genetic algorithm refines model parameters to match morphological experimental observations [6], [7].

The digital twin evolves from software-based simulations accelerated with GPUs to resource-constrained field-programmable gate array (FPGA) implementations. These hardware-compatible versions preserve model fidelity while achieving significant improvements in performance and energy efficiency. Parallel updates inherent to the CA design make FPGA a natural platform for scaling real-time simulations of fungal networks and deploy its low-power digital twin next to the biological counterpart for future real-time interactions.

Extracting the morphological structure of mycelium from FPGA, the paper explores the dynamic electrical activity of fungal tissues. Fungi displays stimulus-responsive electrical potentials that resemble neural spiking, forming the basis for integrating memristive circuits. A memristive oscillating system acting as a cellular automaton is proposed, enabling adaptive signal propagation and local reconfiguration. These circuits are designed to replicate spiking behavior, inhibition, and wave collision dynamics observed in experimental fungal networks, particularly under varying environmental stimuli.

This approach demonstrates a scalable, bio-realistic, and tunable digital twin capable of responding to

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both morphological and electrophysiological experimental data. The findings lay the groundwork for future applications in neuromorphic computing, adaptive sensing, and bio-hybrid system design, marking a step forward in the convergence of biological systems with engineered electronics.

II. Mycelium Morphological Evolution

Mycelial growth is a spatially and temporally dynamic process characterized by decentralized expansion, branching, and interaction with environmental substrates [8], [9]. Capturing these behaviors computationally requires a modeling framework that integrates biological realism [10] with adaptability for hardware deployment. The modeling approach adopted in this work grounded in reaction-diffusion cellular automata (CA), where each cell represents a discrete element of the substrate populated by activator, suppressor, and nutrient variables. The emergent mycelial behavior results from localized interactions among these variables over time.

The CA framework is driven by simple, local rules derived from partial differential equations [10], discretized into finite difference approximations. Activator concentrations promote hyphal tip extension, while suppressor concentrations introduce growth inhibition, reproducing natural branching thresholds and colony morphology. Substrate uptake influences nutrient availability and spatial competition, providing further biological accuracy. Environmental parameters play a crucial role in determining the direction and intensity of fungal growth. Factors such as temperature, humidity, and light were integrated into the simulation by modifying reaction-diffusion coefficients and branching probabilities.

Experimental studies on fungi such as Schizophyllum commune and Trametes versicolor have shown that low-wavelength light (e.g., UV and blue) triggers photomorphogenic responses [11], often resulting in aerial hyphae formation and increased density. In contrast, red and far-red light have minimal impact, closely resembling dark-grown conditions [7]. Temperature and humidity modulate metabolic activity and biochemical reaction rates, with higher values accelerating hyphal extension and biomass accumulation [12], [13].

To align the model with experimental morphological data, a genetic algorithm based in CA was employed for parameter optimization [14]. Morphological descriptors such as branching density, hyphal diameter, and network coverage were extracted from laboratory-grown fungal colonies and compared against simulated outputs. The evolutionary optimization process adjusted CA parameters, including diffusion coefficients, threshold values, and environmental scaling factors. This iterative tuning reduced discrepancies between experimental and simulated structures, allowing the model to replicate key growth patterns observed under various cultivation regimes.

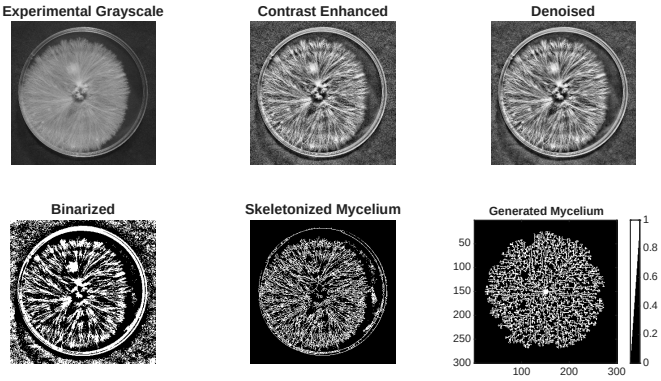


Fig. 1. Experimental photo of mycelium Pleurotus grown on a circular plate under lab conditions and step-by-step processing to export the equivalent skeleton of the structure. The last subplot (bottom-right) indicates the mycelium generated by our reaction-diffusion model.

In Fig. 1 one can see a real laboratory image of Pleurotus mycelium at an advanced evolution stage that can allow us to study its morphological structure. In order to do that we have applied several processing to extract some useful features. The initial image has been converted to grayscale and then we have performed contrast increment and denoising to be ready for the next step, the binarization. At this step the image consists of black and white pixels with the latter signifying the material. After all, we have applied a mask to isolate the information out of the circular growth plane and the Palágyi-Kuba thinning algorithm to extract the material's branches [15]. Having now the growth information the skeleton of the real structure is compared side-by-side with the generated structure from the software model described earlier seen in the bottom-right subfigure showing a plausible replication.

Except that, the photomorphogenic responses have also been studied testing the growth model under various light conditions. Metrics such as the structure of the density and the heterogeneity have been exported and are seen in Fig. 2 showing that the proposed growth model comes to align with the experimental data presented in [7], thus validating that lower wavelengths are more beneficial to the morphological evolution of the material as reported in [11].

III. FPGA Implementation of Mycelium Morphological Models

The FPGA platform was selected as the target digital architecture for implementing the CA-based digital twin due to its inherent parallelism, low power consumption, and reconfigurability. These characteristics align well with the localized and parallel nature of mycelial growth, where each cell's state update depends solely on its immediate neighborhood. The transition from software simulations to FPGA hardware enables real-

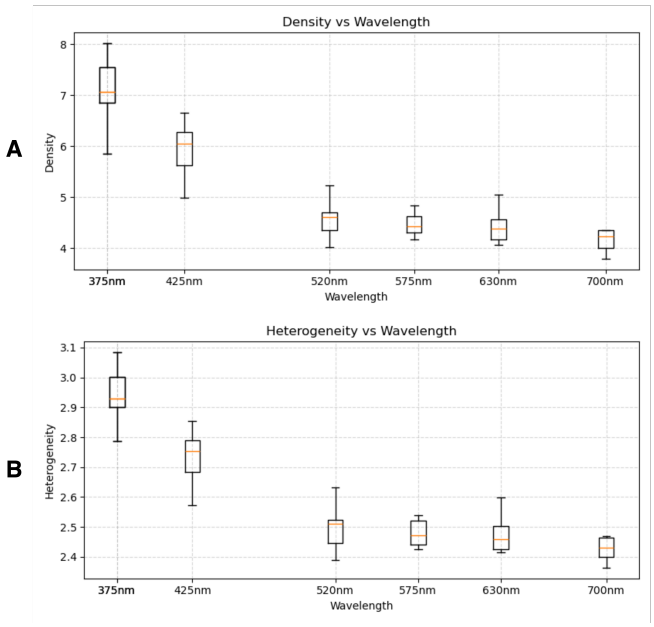


Fig. 2. Box plots of density (A) and heterogeneity (B) of mycelium growth at various light wavelengths showing also the standard deviation after multiple model runs for each wavelength.

time performance while maintaining biological accuracy and morphological complexity.

A central feature of this implementation is the ability to incorporate environmental feedback dynamically. Sensor data for temperature, humidity, and light intensity can be fed into the FPGA in real-time and used to modulate growth parameters at each simulation step. This creates a bidirectional interface between physical cultivation environments and the digital twin, enabling live adaptation of the morphological model to empirical conditions.

Resource optimization was critical to achieving large-scale simulations. The design supports a 100×100 CA grid, with each cell performing three parallel updates for activator, suppressor, and mycelium concentration. Memory access patterns were optimized to allow single-cycle neighborhood data retrieval, and redundant arithmetic operations that were decreasing the performance and efficiency were removed. The resulting design achieves low-latency updates, allowing time-efficient exploration of growth scenarios and morphological adaptation [16].

Validation of the FPGA implementation involved benchmarking the hardware simulation output against the reference software model. Key metrics, including mean squared error (MSE) and root mean squared error (RMSE), were used to compare morphological patterns at various time steps showing a high degree of correspondence between the software and hardware versions, thus providing a robust and extensible tool for simulating mycelial morphology in real time. To be more specific, the use of 16 bits was the more balanced option for resource management and growth modeling errors with the latter to be increasing with each iteration and at the end of growth on a 100×100 grid to have a ratio of $MSE/RMSE = 0.3/0.58 = 0.517$.

IV. Electrical Behavior of Mycelium and Memristor-Based Circuits

While the morphological evolution of mycelium has been well-documented, its electrical behavior, manifested as low-amplitude, stimulus-dependent spiking events, remains a relatively unexplored but highly promising domain for neuromorphic and bio-hybrid applications. Mycelial electrical activity, recorded under diverse stimuli such as heat, chemical exposure, and mechanical pressure, shows patterns analogous to neuronal action potentials [17]. These include solitary spikes, rhythmic oscillations, and signal attenuation upon collision, all of which reflect the organism's capacity for stimulus processing and memory-like behavior.

To emulate these electrophysiological properties, a memristive circuit-based extension of the digital twin has been introduced. At the core of this implementation is a memristive oscillating system (MEMOS) acting as a cellular automaton, a circuit architecture that integrates memristors with localized oscillators to simulate the propagation of electrical signals within fungal networks. Each cell in the MEMOS acts as both a logic element and a memory node, enabling it to adaptively respond to prior activity based on the presence or the absence of the material. This is achieved through voltage-controlled resistance modulation, which encodes historical stimulation in the conductance state of each memristor. The mycelium map on top of that the propagation happens has been generated by the FPGA-based growth model implementation described in the previous Section III.

The MEMOS enables complex spiking dynamics through state transitions governed by electrical thresholds. When a cell receives a stimulus above its threshold, the memristor enters a high-conductance state, triggering oscillations in a coupled oscillator and enabling signal propagation to neighboring cells. This activation spreads in a wave-like manner across the grid and the cell circuit as well as the grid topology can be found in [18]. When two such waves meet, they cancel each other, a phenomenon also observed in Kombucha (tea fermented by a symbiotic community of bacteria and yeasts) [19]. This behavior has been reproduced on the 100×100 simulation grid as one can see in the relevant experiment conducted in [18]. Such behavior mirrors excitable media dynamics and provides a biologically grounded mechanism for signal routing and inhibition. For this experiment, two independently stimulated cells generated electrical waves that propagated across the grid and nullified upon collision.

To align the circuit output with fungal electrophysiology, spike waveforms were compared against recordings

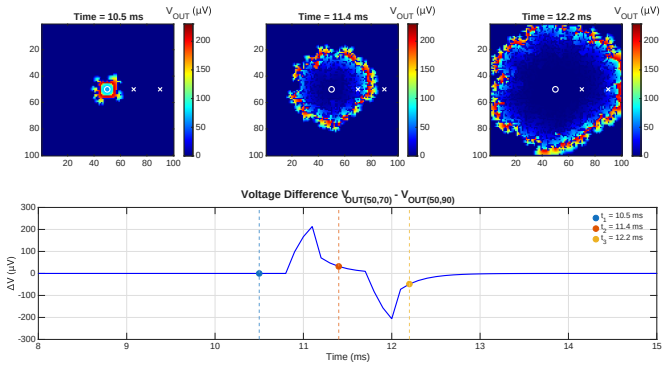


Fig. 3. (top) Snapshots of the electrical activity on the 100×100 grid on a stimuli at the center of the grid (cell is marked with "O"). (bottom) Voltage difference between electrodes in the same row (marked with "X").

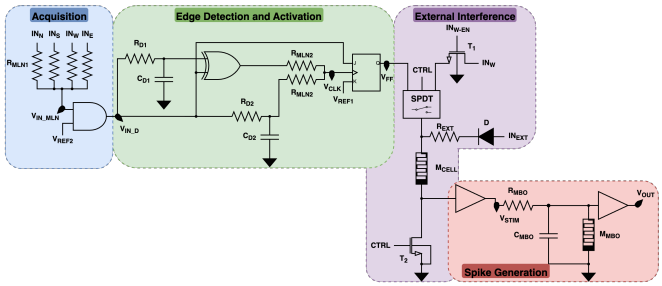


Fig. 4. Circuit diagram of the grid's individual cell.

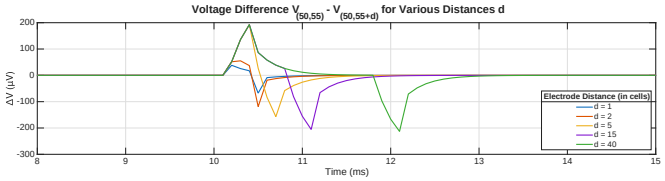


Fig. 5. Voltage difference between an active electrode permanently recording the activity of the cell with coordinates [50, 55] and multiple passive ones recording different cells of the same row which are at a distance d from the active one.

from fungal species such as Schizophyllum commune [20]. These studies revealed consistent action potential-like spikes with identifiable depolarization and repolarization phases, which were replicated in the circuit through phase-specific control of the memristive oscillators. Simulations show that these circuits can reproduce isolated spikes, periodic oscillations, and reactivation following inhibition, aligning with electrical activity recorded in both monocultures and fungal-bacterial co-cultures.

The MEMOS grid also supports reconfigurable topologies. Using memristors to define inter-cell connectivity, the state of each cell can dynamically change based on environmental or activity-based feedback. This flexibility allows the simulation of diverse mycelial architectures and supports dynamic adaptation to changing external conditions. For example, changes in humidity or nutrient levels can trigger topological reconfigurations that mirror hyphal plasticity in vivo.

Circuit-level simulations were conducted using Cadence Virtuoso, validating the spatiotemporal dynamics of the memristive grid utilizing circuit parameters found in [18]. However, there have been modifications at the following parameters in order to enhance the responsiveness of the grid and also to better match the amplitude of the produced waveforms with experimental data from [20], [21]. These parameters are: $C_{MBO} = 100$ pF, $C_{D2} = 300$ nF, $R_{MLN1} = 10$ kΩ, while now for the external stimulation a pulse with amplitude equal to 0.25 V and a duration of 150 μs is being used meaning that with the new parameters the grid now is more responsive. Also, the circuit of the grid's cell with the distinct processing stages that have been described in [18] is now seen in Fig. 4 slightly modified as the V_{STIM} voltage that drives the memristive oscillator comes from an operational amplifier (op-amp) and not from the threshold-based AND gate as reported in [18]. This gives the advantage of the possible environmental contribution through the IN_{EXT} pin not only to the electrical activity of the cell (for instance its firing rate) but also to its long-term state (if it is able to transmit a signal or not).

Voltage difference between two grid points is a crucial metric to evaluate the electrical activity of mycelium. As seen in Fig. 3 (bottom) the voltage difference between a pair of electrodes on the proposed grid appears dynamics that are also met in differential recordings of S. commune keeping the amplitude of the produced spiking voltages V_{OUT} at realistic levels (hundreds of μV). Other than that, the voltage difference has been studied in multiple cases in which the distance of the differential electrodes was varying. In Fig. 5 the different distance between the two electrodes is well captured at the voltage difference waveforms as the ΔV amplitude and the timing between the peaks are different in every test case replicating the behavior met in [20].

In a more macroscopic view, mycelium can appear more complex electrical activity that emerges over time in multiple stimuli. In order to replicate the activity not in a standalone triggering event but in a bunch of external stimuli, external signals have been applied at random cells of the grid, that represent material and can transmit information. The triggering events at first have a time difference of $\Delta t = 2$ ms and it gradually decreases so that the last stimuli have $\Delta t = 0.25$ ms for the evaluation of the responsiveness of the grid which performs well. The corresponding results are seen in Fig. 6.

Power analysis revealed that while peak power consumption reached 3.16 W during maximal activity, the total energy usage remained low due to the non-volatile nature of memristive elements. The circuit preserved prior activity patterns even after power cycles, supporting long-term memory functionality essential for adaptive

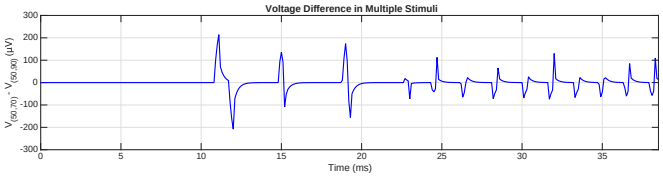


Fig. 6. Voltage difference between an active electrode recording the activity of the cell with coordinates [50, 70] and a passive one recording the activity of another cell located at [50, 90] in external stimuli applied in random cells.

and reservoir computing.

Through this hardware-software hybrid system, the electrical behavior of mycelium is no longer an abstract biological phenomenon but a directly emulatable, tunable feature. By bridging fungal electrophysiology and reconfigurable electronics, the framework paves the way for self-organizing, low-power, and learning-capable biohybrid systems.

V. Conclusions and Future Work

This work introduces a hardware framework for modeling and implementing the morphological and electrical behavior of mycelium, combining reaction-diffusion cellular automata with real-time hardware platforms. The digital twin captures key fungal features, such as tip extension, branching, and adaptive substrate interaction, while incorporating environmental parameters like temperature, humidity, and light. Evolutionary optimization aligns simulated outputs with experimental morphological data, enabling biologically accurate and hardware-compatible modeling suitable for real-time applications.

The framework is extended through the integration of memristor-based circuits, forming a memristive oscillating system (MEMOS) that replicates the electrical spiking activity observed in fungi. This architecture supports reconfigurable signal propagation, non-volatile memory, and spatiotemporal computing dynamics, laying the foundation for neuromorphic and bio-hybrid systems. Guided by experimental feedback, including co-cultivation datasets, the platform offers a scalable and adaptive pathway toward low-power, responsive engineered living materials that reflect the computational potential of fungal networks.

Future work will focus on extending the digital twin architecture into aggregate-level systems capable of learning from multi-sample fungal cultures. The integration of adaptive environmental sensing and bio-signal acquisition circuits with hardware will support dynamic reconfiguration and long-term operation in complex environments. Additionally, coupling the digital twin with machine learning algorithms will open avenues for predicting emergent mycelial behavior and optimizing living material properties under diverse operational conditions.

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